Ansible Playbook Run Results

1. Basic Example of Ansible Output (successful tasks)

```
bash

$ ansible-playbook example_playbook.yml

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
*********************************************************
ok: [webserver1]

TASK [Install Nginx]
*********************************************************
changed: [webserver1]

TASK [Start Nginx service]
*****************************************************
ok: [webserver1]

PLAY RECAP
************************************************************************
webserver1                 : ok=3    changed=1    unreachable=0
failed=0    skipped=0    rescued=0    ignored=0
```

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Explanation:

- **ok:** The task was successfully completed without changes.
- **changed:** The task made changes on the target host.
- **unreachable:** The host could not be reached.
- **failed:** The task failed.
- **skipped:** The task was skipped based on a condition.
- **rescued:** Tasks that were rescued after a failure.
- **ignored:** Any task errors were ignored using `ignore_errors: yes`.

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2. Output with a Failed Task

```
bash
$ ansible-playbook example_playbook.yml

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
************************************************************
ok: [webserver1]

TASK [Install Nginx]
************************************************************
failed: [webserver1] (item=nginx) => {"changed": false, "msg": "Could not
find the Nginx package", "item": "nginx"}

PLAY RECAP
********************************************************************
webserver1                  : ok=1    changed=0    unreachable=0
failed=1    skipped=0    rescued=0    ignored=0
```

Explanation:

- The task failed due to an error (in this case, Nginx package not found).
- The output highlights the error message (`msg`) and the failing task (`item`).

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3. Output with Skipped Tasks

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```
bash
$ ansible-playbook example_playbook.yml

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
************************************************************
ok: [webserver1]

TASK [Install Nginx]
************************************************************
skipping: [webserver1]

TASK [Start Nginx service]
************************************************
ok: [webserver1]

PLAY RECAP
********************************************************************
webserver1                  : ok=2    changed=0    unreachable=0
failed=0    skipped=1    rescued=0    ignored=0
```

Explanation:

- The second task (`Install Nginx`) was skipped, often because of a `when` condition, and it's marked as **skipped**.

4. Unreachable Hosts

```
bash

$ ansible-playbook example_playbook.yml

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
*********************************************************
fatal: [webserver1]: UNREACHABLE! => {"changed": false, "msg": "Failed to
connect to the host via ssh: Connection timed out", "unreachable": true}

PLAY RECAP
********************************************************************
webserver1                  : ok=0    changed=0    unreachable=1
failed=0    skipped=0    rescued=0    ignored=0
```

Explanation:

- An **unreachable** host error occurs when Ansible cannot connect to the host (e.g., due to network issues, SSH issues, etc.).
- The `UNREACHABLE!` status appears, and the task shows what went wrong (e.g., `Connection timed out`).

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5. Output with Verbosity (`-v`, `-vv`, `-vvv`)

If you increase the verbosity with `-v`, `-vv`, or `-vvv`, you'll see more detailed output, such as the exact commands being run on the remote host and more detailed task results.

```
bash

$ ansible-playbook example_playbook.yml -v

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
*********************************************************
ok: [webserver1] => (item=None) => {
    "ping": "pong"
}

TASK [Install Nginx]
*********************************************************
changed: [webserver1] => (item=nginx) => {
    "changed": true,
    "msg": "Package nginx installed"
}
```

```
TASK [Start Nginx service]
*************************************************
ok: [webserver1] => (item=None) => {
    "service": "nginx",
    "state": "started"
}

PLAY RECAP
*********************************************************************
webserver1                 : ok=3    changed=1    unreachable=0
failed=0    skipped=0    rescued=0    ignored=0
```

Explanation:

- The `-v` option provides more detail, including task parameters and module results.
- It can be useful for debugging tasks or understanding what's happening under the hood.

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6. Colorized Output (Default)

By default, **Ansible** produces colorized output (if the terminal supports it) for clarity:

- **Green:** Task completed successfully.
- **Yellow:** Task resulted in a change.
- **Red:** Task failed.
- **Grey:** Skipped or informational tasks.

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Example:

```
bash
$ ansible-playbook example_playbook.yml
PLAY [Web Servers]
*************************************************************
TASK [Ping web server]
*************************************************************
ok: [webserver1]
TASK [Install Nginx]
*************************************************************

changed: [webserver1]

TASK [Start Nginx service]
*************************************************
ok: [webserver1]

PLAY RECAP
*********************************************************************
webserver1                 : ok=3    changed=1    unreachable=0
failed=0    skipped=0    rescued=0    ignored=0
```

In a colorized terminal:

- The `ok` tasks would show in **green**.
- The `changed` task would show in **yellow**.

7. Using `--check` for Dry Run (No Changes)

When running with the `--check` flag (dry run), no changes are made, but you can see what would happen:

```
bash
$ ansible-playbook example_playbook.yml --check

PLAY [Web Servers]
************************************************************

TASK [Ping web server]
*********************************************************
ok: [webserver1]

TASK [Install Nginx]
*********************************************************
ok: [webserver1]

TASK [Start Nginx service]
*****************************************************
ok: [webserver1]

PLAY RECAP
************************************************************************
webserver1                 : ok=3    changed=0    unreachable=0
failed=0    skipped=0    rescued=0    ignored=0
```

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Explanation:

- No changes were made because of the `--check` flag, but it will still show what would have been done.

Summary of Key Output States:

1. **ok** – Task completed successfully, no changes.
2. **changed** – Task completed with changes made to the target.
3. **failed** – Task failed, details are included.
4. **skipped** – Task skipped due to conditions.
5. **unreachable** – Host could not be reached.